Applied Machine Learning for Business

Course Project – Interim Submission
Delivery Delay Prediction for Brazilian E-Commerce

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1 Problem and Decision

Decision-maker. Operations Manager at **Olist**, a Brazilian e-commerce marketplace (27 states, 4,119 cities).

The problem. Of 96,476 delivered orders (Sep 2016 – Oct 2018), 7,827 (8.1%) arrived after the platform's estimated delivery date. These late deliveries correlate with a 19× increase in poor reviews (Figure 2), damaging platform trust, seller retention, and customer lifetime value. During Q2 (April–June), order volume surges 11%, straining logistics and amplifying delay risk.

Decision. At the moment an order is placed before shipping begins should the operations team flag it for *Priority Handling* (expedited warehouse processing, premium carrier, proactive seller notification)? The data shows 7,827 orders needed intervention over 26 months; the model's job is to identify them at placement time so the team can act before delays materialize.

Economic assumptions. We assume: (i) each late delivery costs **R$150** in reputation damage, churn risk, and reduced future orders; (ii) priority handling costs **R$20** per order (carrier upgrade R$10, expedited processing R$7, overhead R$3); (iii) standard logistics costs R$5 per order. These are constructed assumptions we do not have Olist's actual cost structure.

	Predicted On-Time	**Predicted Late**
Actually On-Time	R$5 (standard cost)	R$20 (wasted premium)
Actually Late	**R$155** (R$150 loss + R$5 standard)	R$20 (premium prevents loss)

Table 1: Confusion cost matrix (assumed). A false alarm (FP) costs R$20; a missed late delivery (FN) costs R$155. The ratio R$155 / R$20 ≈ **8:1** drives evaluation toward catching late orders over avoiding false alarms.

Evaluation metric: Total Cost = FP × 20 + FN × 155, using the costs directly from Table 1. We optimize the decision threshold to minimize this cost, *not* accuracy.

2 Data

Source. Olist Brazilian E-Commerce Public Dataset (Kaggle), publicly available for academic use. **Period.** September 2016 – October 2018 (26 months). **Access confirmed:** all tables loaded.

The dataset comprises **8 relational tables**. The `Orders` table is the central hub: each order links to its items via `Order Items` (using `order_id`), which in turn references `Products` (via `product_id`) and `Sellers` (via `seller_id`). Customer geography joins through `Customers` (via `customer_id`),

and both customer and seller zip codes map to lat/long coordinates via the `Geolocation` table. `Payments` and `Reviews` attach to orders directly via `order_id`.

Table	Records	Key Fields
Orders	99,441	timestamps, status
Order Items	112,650	price, freight, seller_id
Customers	99,441	city, state, zip
Products	32,951	category, weight, dimensions
Payments	103,886	type, installments, value
Reviews	99,224	score (1–5), comments
Sellers	3,095	city, state, zip
Geolocation	1,000,163	latitude, longitude

Table 2: Dataset structure. Tables join via `order_id`, `customer_id`, `product_id`, `seller_id`.

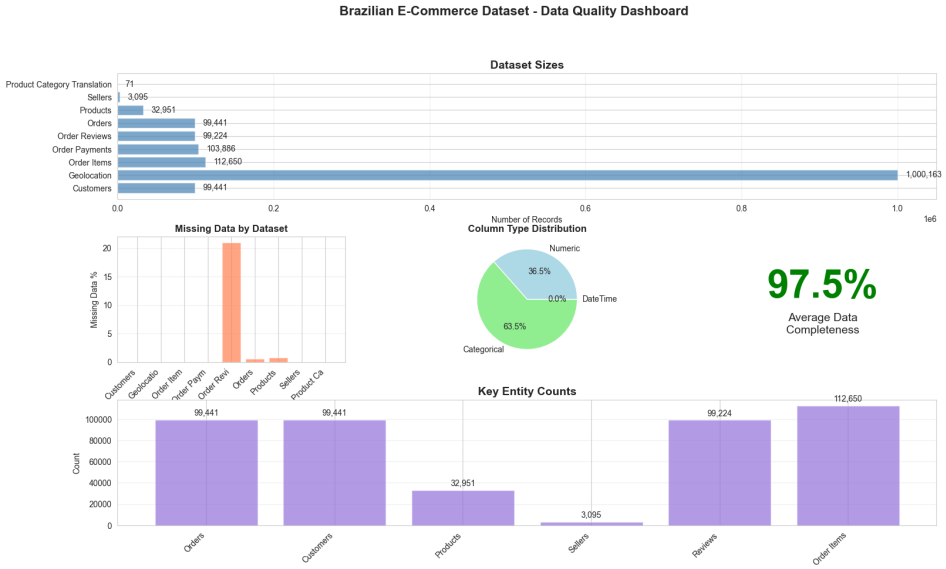


Figure 1: Data quality dashboard confirming 97.5% average completeness across all 8 tables, with dataset sizes and column type distribution.

Target variable (derived): We construct `is_late` from the raw timestamps in the `Orders` table: `is_late` = 1 if `order_delivered_customer_date` > `order_estimated_delivery_date`; 0 otherwise. This column does not exist in the raw data and is engineered during preprocessing.
Distribution: 88,649 on-time (91.9%) vs. 7,827 late (8.1%). Class imbalance ratio: 1 : 11.3.

Key categorical variable: The `Products` table contains 74 distinct product categories (e.g., bed_bath_table, health_beauty, sports_leisure, computers_accessories, furniture_decor). Categories vary significantly in late delivery risk, with bulky/heavy categories like furniture showing higher delay rates than lightweight categories like telephony.

Sample data (8 representative orders from the dataset):

Cust.	Seller	Category	Price	Freight	Wt (g)	Delay (d)	Review
SP	SP	housewares	R$30	R$9	1,200	−2.5	5
BA	SP	perfumery	R$119	R$23	850	+1.3	3
RS	SC	auto	R$160	R$19	3,500	−5.8	5
RN	SP	pet_shop	R$45	R$27	2,100	+8.2	1
MG	MG	bed_bath	R$90	R$15	4,200	−1.2	4
SP	SP	sports	R$199	R$12	1,800	−6.5	5
CE	PE	telephony	R$380	R$32	650	+4.7	2
PR	SP	furniture	R$549	R$46	18,500	+12.3	1

Table 3: Sample orders. Positive delay = late. Late orders cluster in cross-state, heavy-item routes.

3 Summary Statistics

Missing data. Core features are 97% complete. Missing delivery dates (2,965 orders, 3%) correspond to undelivered/canceled orders and are excluded. Product category is missing for 610 items (1.9%, imputed as "unknown"). All other fields are fully populated.

Non-trivial finding: delivery delay drives poor reviews exponentially. Figure 2 shows the poor-review rate (1–3 stars) rises from 19.5% for on-time orders to 85% for orders 8–14 days late a **4.4× increase**. This is the strongest single predictor of customer dissatisfaction and validates our target variable choice.

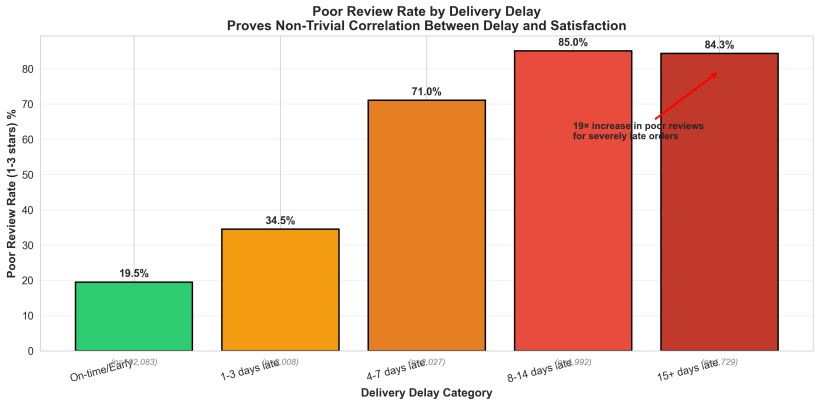


Figure 2: Poor review rate by delivery delay category. The steep, monotonic increase confirms that delivery delay is the dominant driver of customer dissatisfaction.

Geographic routing matters. Cross-state deliveries (55.6% of orders) have a late rate of 9.0% vs. 6.0% for same-state a **1.5× gap** (Figure 3). São Paulo accounts for 80.7% of seller-side orders but only 42% of customer-side, meaning most shipments travel long distances.

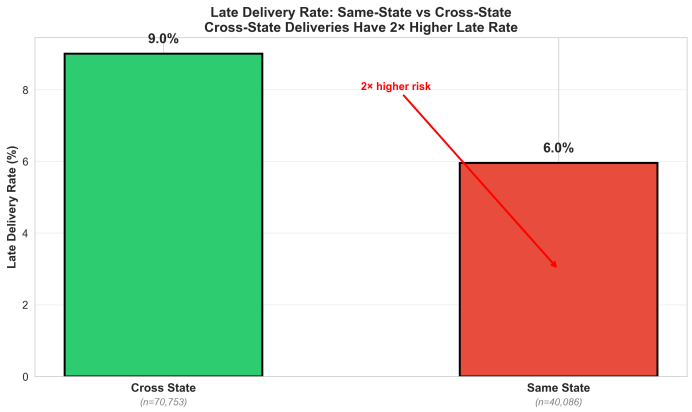


Figure 3: Late rate by routing type. Cross-state deliveries carry notably higher delay risk than same-state shipments.

4 Proposed Approach

Task: Binary classification (`is_late`, derived from delivery timestamps), predicted at order placement (before shipping).

Features (51 total, 9 groups):

- **Geographic (6):** Haversine distance (km), same-state/city flags, state-pair historical late rate expected strongest group.
- **Product (13):** Weight, volume, bulky-item flag, category, category-level late rate.
- **Temporal (8):** Month, quarter, weekend flag, peak-season indicator (Q2).
- **Order (6):** Item count, total value, freight cost, freight-to-value ratio (rural proxy).
- **Payment (7):** Type (one-hot), installments, value.
- **Seller (3):** Historical order count, late rate, state volume.
- **Time pressure (3):** Estimated delivery window, urgent/long-window flags.
- **Interactions (5):** distance×weight, distance×bulky, peak×distance to capture compounding risk.

Models: Logistic Regression (baseline), Random Forest, and XGBoost (primary). Tree-based models handle high cardinality (74 product categories, 27 states), capture non-linear interactions, support class weighting, and provide feature importance for business interpretation.

Key challenge: class imbalance (8.1% late). We address this via (i) class weights (11× minority), (ii) SMOTE oversampling, and (iii) threshold tuning. Because 91.9% of orders are on-time, a model that blindly predicts "on-time" for every order would achieve 92% accuracy while catching zero late orders. This accuracy paradox is why we evaluate on cost, not accuracy.

Evaluation: We minimize Total Cost $= \text{FP} \times 20 + \text{FN} \times 155$ (Table 1), not accuracy. The optimal threshold is expected below the default 0.5, reflecting the 8:1 cost asymmetry. Secondary metrics: Recall (priority), F1, ROC-AUC.

Train/test split: We will evaluate using both (i) an **80/20 random split** for standard generalization testing, and (ii) a **temporal split** (train on 2016–2017, test on 2018) to simulate real-world deployment where the model predicts future orders from past data. Comparing both strategies will reveal whether temporal patterns affect model reliability.

Sensitivity analysis: We will test robustness across four cost scenarios (review loss R$100–300, intervention cost R$20–40) to verify that the core insight loss avoidance dominates survives reasonable perturbation of assumptions.

References

[1] Olist Brazilian E-Commerce Public Dataset. Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce